

# Tokenization in Gaming Industry



Instructed by  
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# Agenda

- 01** Introduction & The Core Problem
- 02** From Game Items to Tokenized Assets
- 03** Economic Models & Sustainability
- 04** Case Studies: Evidence from the Industry
- 05** Challenges, Evaluation & Insight

# 01 Introduction & The Core Problem

Market Opportunity

# Gaming is Enormous



TAM • Total Market

**\$201.6B**

All global video game software and hardware spending.

SAM • Virtual Goods

**\$103.5**

Cosmetic items, **B** battle passes, and virtual economies.

SOM • On-chain Gaming

**\$36.19B**

Immediate market addressable by tokenized infrastructure.

# Introduction & The Core Problem

## Industry Composition

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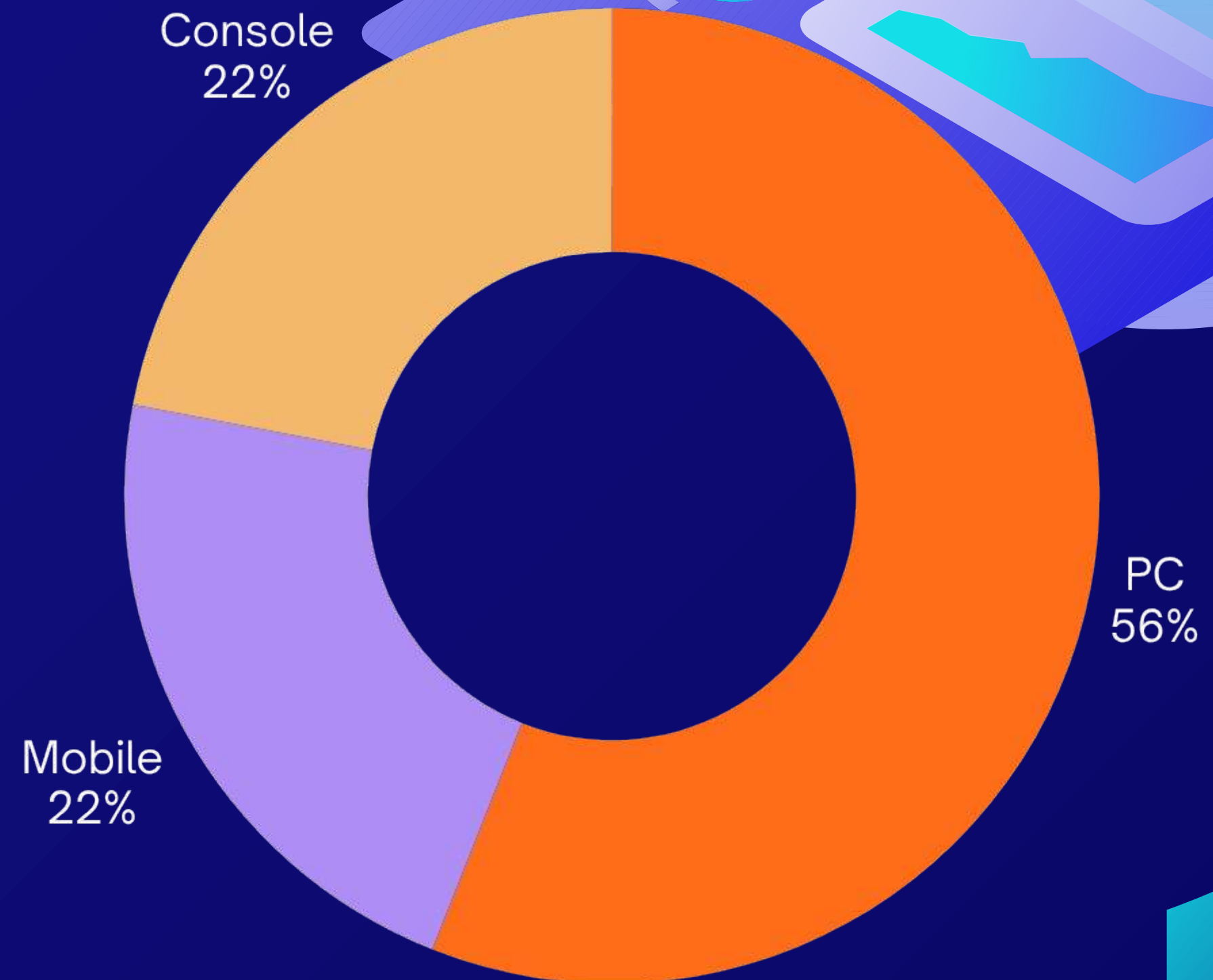
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# Introduction & The Core Problem

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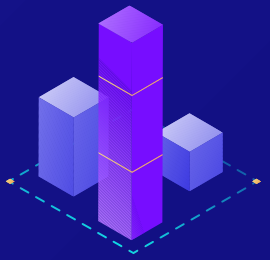
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# Introduction & The Core Problem

## Consumer Shift



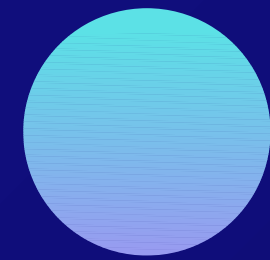
**Live-Service Pivot:** Monetization shifted permanently from retail discs to continuous micro-purchases.



**The Spectrum of Virtual Assets:** Modern game revenue is driven by cosmetic character skins, weapon wraps, emotes, battle passes, and virtual land parcels.



**Fiat Inflow:** Billions converted into non-refundable, closed in-game currency points (e.g., V-Bucks, Riot Points, Robux).



# Introduction & The Core Problem

The Core Problem: The Illusion of Digital Ownership

“

I bought a skin for \$500.

Do I actually own it?

---

Under modern Terms of Service, players hold a temporary, conditional license. Publishers retain unilateral rights to alter, lock, or erase the asset.

# Introduction & The Core Problem

Systematic Vulnerabilities

## Walled Gardens

Zero asset  
exportability across  
external platforms.

## Server Depreciation

Decommissioned  
game servers delete  
player collections.

## Arbitrary Revocation

Account bans  
permanently forfeit  
all digital inventory.





## Introduction & The Core Problem

# Tokenization

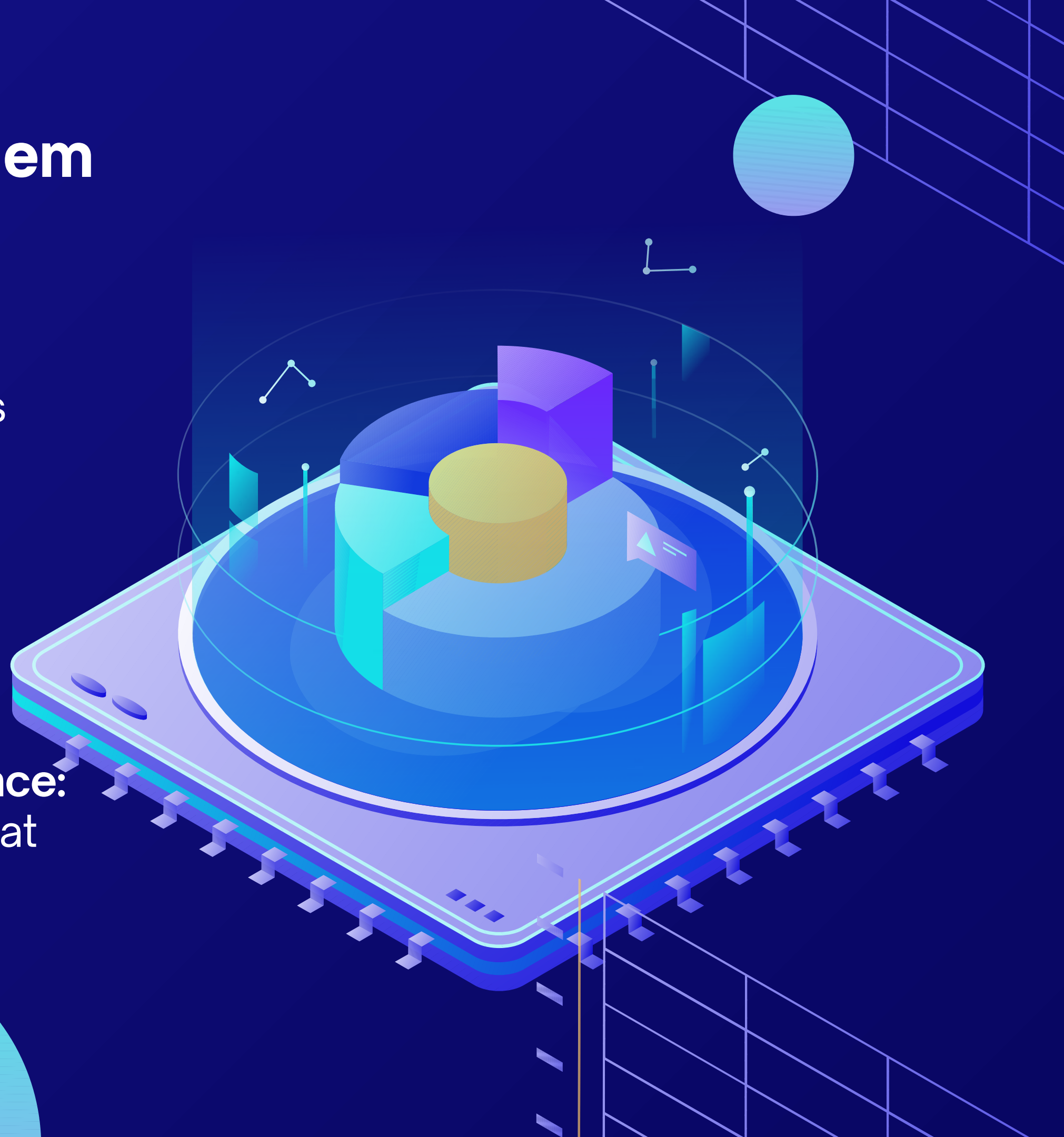
Converting in-game cosmetics, gear, and currencies into on-chain cryptographic tokens (NFTs/ERC-721/ERC-1155) stored in player-custodied wallets.



# Introduction & The Core Problem

## Tokenization

- 01 Player Wallet Custody:** The player holds private cryptographic access keys.
- 02 Programmable Smart Contracts:** Game items mapped to on-chain non-fungible tokens.
- 03 Transparent & Auditable Provenance:** Permanent on-chain provenance that guarantees verifiable scarcity.



# Wrap Up

We are living in a paradox.



## Economic Value vs. Legal Reality

\$103B Virtual Economy - Huge spending is driven by status, but legally, players only hold a revocable license.



## The Architectural Constraint

Walled Garden Bottleneck - Items live on private servers; bans or shutdowns erase collections with zero recourse.



## Tokenization as the Proposed Shift

Tokenized Alternative: Public ledgers and wallet custody decouple assets from publisher databases.



# 02 From Game Items to Tokenized Assets

## Tokenization

How blockchain changes ownership, trading, and the value of digital game assets





# From Game Items to Tokenized Assets

## Traditional vs. Tokenized Assets

Traditional gaming relies on publisher databases, while tokenized gaming uses blockchain-based tokens.

### Traditional Gaming (Web2) - Centralized



- Publisher-managed access/publisher.
- Ecosystem-locked assets.
- Platform-controlled trading.

### Tokenized Gaming (Web3) - Decentralized



- Tokenized game assets.
- Blockchain-verified control.
- Transferable and tradable.



**Key Takeaway:** Traditional gaming depends on a publisher-controlled database; tokenization creates a blockchain-based ownership record.

# From Game Items to Tokenized Assets

What Value Does Tokenization Add?

Beyond storing an item, blockchain can create new ways to verify, transfer, and manage digital assets.

## 1 Verifiable Ownership & Provenance



### Transaction History

-  Created (2022-01-10)
-  Sold to A (2022-06-15)
-  Transferred to B (2023-03-20)
-  Sold to C (2024-01-12)



You can verify who owns the item, and trace its full history.

## 2 Transferability & Secondary Markets

Player A



Player B



**Example:**  
Player A → sells NFT skin → Player B

## 3 Programmability

95%

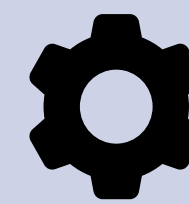


**Seller**  
(eg. Player)

5%



**Developer**  
(royalties)



Automates developer royalties on every secondary sale.

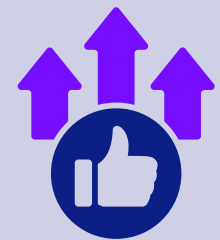


**Key Takeaway:** Tokenization can turn game assets into transferable and programmable digital assets.

# From Game Items to Tokenized Assets

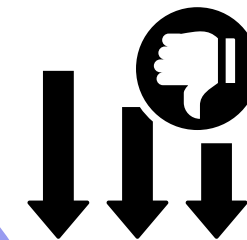
## Advantages vs. Disadvantages

Blockchain creates new possibilities, but it also adds complexity that traditional gaming may not have.



### Advantages

- Verifiable ownership/control
- Transferable digital assets
- Transparent transaction history
- New secondary markets
- Programmable transactions



### Disadvantages

- More technical complexity
- Wallets and transaction fees
- Security risks
- Not useful for every game
- Can introduce speculation



**Key Takeaway:** Tokenization is not automatically better; its benefits must justify the additional costs and complexity.

# Wrap Up

It creates new possibilities but also introduces new costs and risks.



## Traditional Gaming

- Publisher-controlled database
- Limited control and transferability



## Tokenized Gaming

- Blockchain-based token
- Verifiable ownership
- Transferability
- Programmability



## Disadvantages

- Wallets & transaction fees
- Security risks
- Speculation
- Not useful for every game



**Key Takeaway:** Tokenization can add value when the benefits of ownership and transferability outweigh the additional costs and complexity.



# 03 Economic Model & Sustainability

## Tokenization

Analyzing the macroeconomic engine that determines whether a decentralized virtual world survives or collapses.



### Monetary Openness

Asset sovereignty vs. external DEX contagion



### The P2E Death Loop

Extractive design and liquidity runs



### Faucet & Sink Law

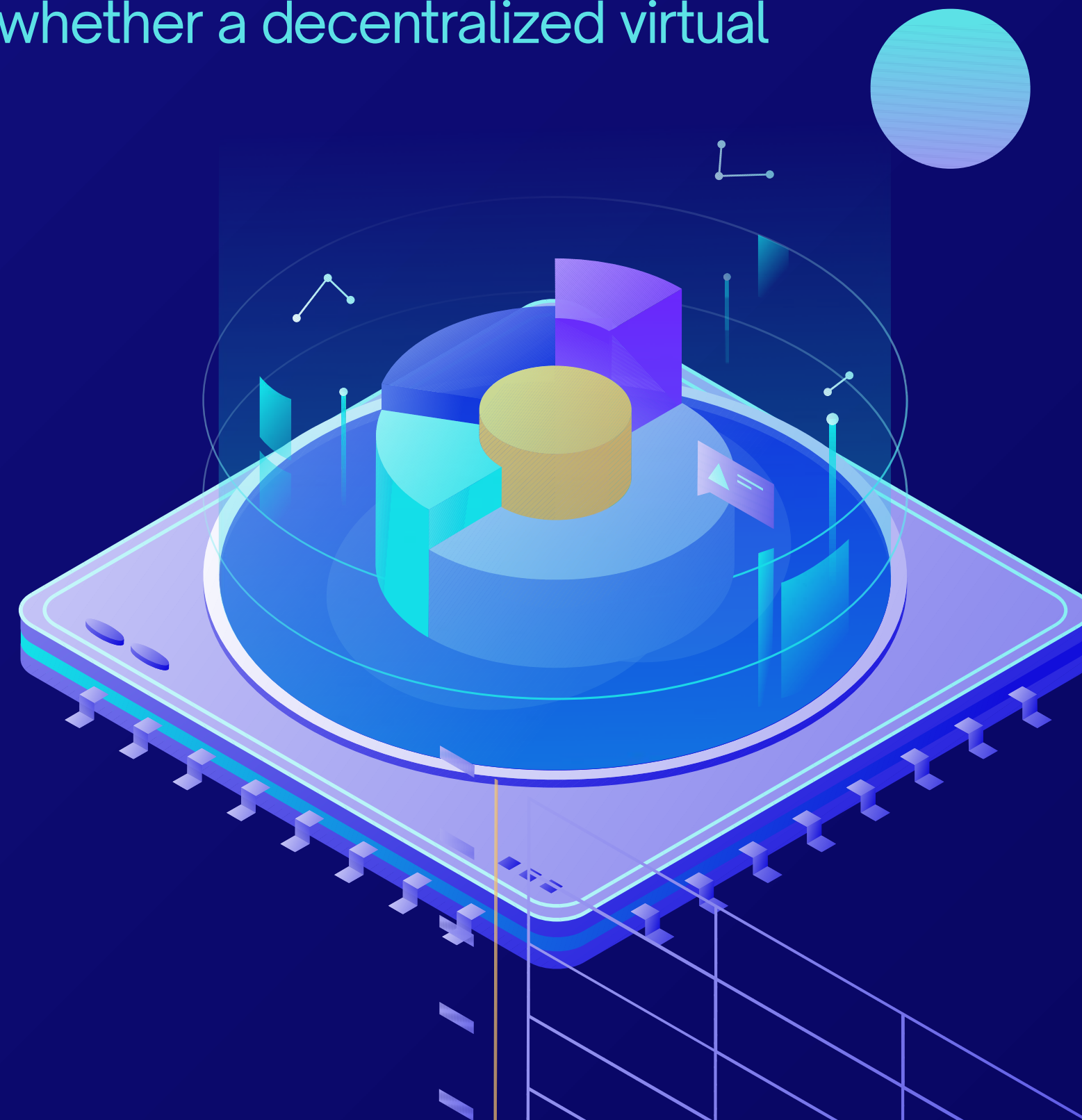
Equilibrium between emissions and burns



### Resilient Blueprint

Anti-sybil and the 6-step roadmap

→ A game cannot survive on speculative token yields; longevity requires positive net consumer demand.



# Economic Model & Sustainability

From Asset Ownership to Economic Models



How tokenization & monetary openness fundamentally redefine the game economy.

## Earn Asset

Gameplay directly generates verifiable liquid tokens and NFT items



## Trade Asset

Secondary peer-to-peer markets trade without platform restrictions



## Unbounded Capital Flow

Tokens continuously enter and leave the perimeter of the game's internal economy.

## Why Web3 Game Economies Are Different?

### External Interaction Vectors

- Exchanges: Instant CEX/DEX spot trading & pricing.
- Secondary Markets: Open, unpermissioned NFT trade.
- External Liquidity: Unrestricted crypto capital pools.

→ Real economic incentives & user ownership

### Systemic Exposure Risks

- Volatility: Market swings break game costs
- Speculation: Traders crowd out real players
- Outflows: Capital drained during downtrend

→ Vulnerability: Gameplay exposed to global crypto cycles



**Key Takeaway:** Economic openness makes Web3 games uniquely powerful, but inherently fragile without strict liquidity control.

# Economic Model & Sustainability

Play-to-Earn (P2E): Financialization First

How early P2E mechanics functioned and why financial incentives replaced gameplay.



## The P2E Cash-out Flow



### Gameplay Activity

Quests, PvP battles, ladder progression



### Token Emission

Minting fungible tokens and NFT drops



### External Cash-Out

DEX swaps for USDT and fiat off-ramps

## Fatal Structural Bottlenecks



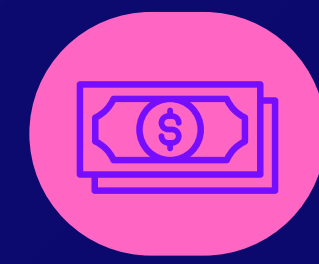
### Money as Product

Players optimize purely for yield; repetitive grinding and bot farms destroy entertainment.



### Extractive Structure

Senior player payouts rely on starter-pass buys from new entrants rather than organic demand.



### Open-Market Run

Uncapped external DEX liquidity triggers collective bank runs at the first sign of slowing growth.

→ A game cannot sustainably create economic value simply by paying players to play.

# Economic Model & Sustainability

The P2E Death Spiral and the Paradigm Pivot to Play-and-Own model

The algorithmic mechanics of collapse and the evolution toward Play-and-Own (P&O).

## The Feedback Loop of Collapse of P2E

1. User Inflow Slows
- 
2. Supply Exceeds Demand
- 
3. Token Dumps on DEX
- 
4. Real Yields Evaporate
- 
- **MASS PLAYER CHURN**

## Play-to-Earn

**“Play to earn money”**

Financial rewards are the core product; retention instantly breaks once payouts decline.

## Play-and-Own

**“Play for fun; own as an added benefit”**

Ownership enhances status, utility, and competition without an hourly wage promise.

**Core Realization:** Retention must be driven by intrinsic gameplay entertainment, never by speculative financial subsidies.



# Economic Model & Sustainability

The Core Engine: Faucets vs. Sinks

Balancing currency creation with currency consumption to prevent hyperinflation.

## Faucets (Supply Inflows)



### Engagement Rewards

Daily quests & progression drops



### Competitive Pools

PvP prize pools & tournament loot



### Gear Drops

RNG item minting & exploration loot

**EQUILIBRIUM  
LAW**

$$\Delta \text{ Supply} = \text{Faucets} - \text{Sinks}$$

## Sinks (Supply Outflows & Locks)



### Flow Sinks (Permanent Burns)

- Crafting, breeding fees, & item durability repairs
- Marketplace protocol burns & event entry fees



### Stock Sinks (Capital Locks)

- Governance staking & seasonal escrow
- Guild war treasuries & tier memberships

**Sustainability holds when:**

Consumer Value Injection  $\geq$  Net Liquidity Extraction

# Economic Model & Sustainability

## Token Architecture: Choosing the Right Structure

Does the game truly need an on-chain token, or can off-chain points execute the same function?



### Single-token model

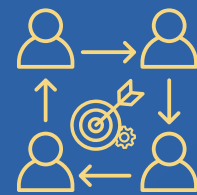
One token for in-game spending and external speculation. Surges make gameplay expensive; crashes destroy incentives.



### Dual-token model

Separates fixed governance from utility currency. However, reward tokens still hyperinflate if burns don't match emissions.

## The 3-Stage Token Necessity Filter



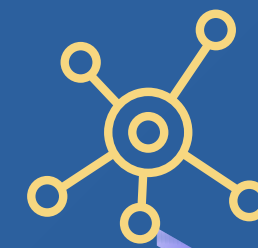
### Consensus Need

Does this currency require permissionless consensus, or do database points suffice?



### Volatility Shield

Will exposing game loops to DEX speculation enhance or ruin core player experience?



### Minimal Complexity

Deploy the simplest architecture that fulfills real gameplay utility.

The objective is not to maximize tokens, but to use the simplest setup that supports actual player needs.

# Economic Model & Sustainability

## Systemic Risks & Common Design Pitfalls

What fatal vulnerabilities repeatedly derail decentralized game economies?

### Systemic Macro Risks

#### Token Inflation

Emissions outpace burns; rewards devalue rapidly.

#### Speculative Bubbles

Yield-chasers push prices far above organic game utility.

#### Botting & Sybil

Automated scripts extract faucets with zero friction.

#### Over-Financialization

Game actions feel like high-stress labor, inducing churn.

### Critical Design Failures

1. Yield-Only Demand

1. Retrofitted Sinks

1. Hyper Emissions

1. Zero Telemetry

1. Late Detection



**Key Takeaway:** Sustainability must be architected into the game from day one; it cannot be patched after token collapse.

# Economic Model & Sustainability

## The Sustainable Blueprint & 6-Step Roadmap

A proven framework for engineering durable, anti-extractive Web3 virtual worlds.



### Fun Gameplay First

Creates genuine willingness to spend for entertainment, vanity, and social status rather than financial yield.



### Skill-Based Rewards

Eliminates passive time-farming payouts. Rewards valuable mastery and competitive contribution.



### Decouple Governance

Isolates on-chain ecosystem governance equity from day-to-day gameplay utility currencies.

## The 6-Step Execution Protocol

Player  
Demand

Necessity  
Test

Sinks  
First

Match  
Faucets

Taper  
Rewards

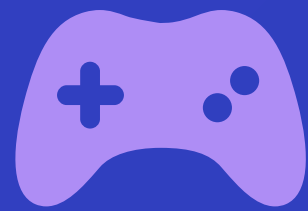
Adversarial  
Test

→ Tokens must serve as infrastructure for great gameplay, not function as the game itself.



# Wrap Up

The 3 core pillars of enduring Web3 titles.



## Fun Gameplay

Generates  
genuine consumer  
spending demand



## Sustainable Tokenomics

Balances token  
emissions against  
real sinks



## Blockchain Infrastructure

Enables trustless  
ownership and  
transferability



**Final Takeaway:** The goal is not to maximize financialization but to build an economy where the game creates enough real value to sustain itself.

# 04 Case Study: Evidence from the Industries

Axie Infinity and Counter Strike



What's keep a game economy alive?

Play-To-Earn



Play-and-Own



# Case Study: Evidence from the Industries

## Axie Infinity Case

Axie Infinity was developed by the Vietnamese studio Sky Mavis in 2018 and trade NFT creatures on Ronin sidechain.



### How to play?



#### Gameplay Activity

Quests, PvP battles  
Pokemon-like gameplay  
\$400 starter pack



#### Faucet

Minting SLP and  
AXS



#### Sink

Consume SLP and  
AXS to breed Axies

### Mid-2021

- 1 AXS = \$165
- 1 SLP = \$0.39
- 2.7M users daily
- > \$4 billion in volume

# Case Study: Evidence from the Industries

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### Mid-2021

- 1 AXS = \$165
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- > \$4 billion in volume

### Mid-2022

- 1 AXS = \$8.5
  - 1 SLP = >\$0.004
  - 95% of players left
- Players are profit-driven

# Case Study: Evidence from the Industries

## Counter Strike 2 Case

Famous FPS game developed by Valve and distribute through Steam.  
#1 Game on Steam with 800K active player.



### Market Cap

- About \$5.9B item economy
- More than 1.4mil concurrent players
- Valve control supply and earn 15% commission

### ⚡ The trade-up rule change

5 skins = 1 knife/glove skin → knife/gloves will be common  
→ the market loses 30% value within 48 hours → the market recovers to about 75-80% of pre-crash levels by early 2026.

# Wrap Up

|           |  <b>Axie Infinity</b> |  <b>Counter Strike 2</b> |
|-----------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Incentive | Play                                                                                                    | Earn                                                                                                        |
| Demand    | ~800K active players                                                                                    | Speculators, new entrants                                                                                   |
| Ownership | Distributor-held (Valve, Steam)                                                                         | On-chain, player-held                                                                                       |
| Outcome   | Volatile but durable                                                                                    | ~99% value collapse                                                                                         |



**Key Takeaway:** Tokenization is not automatically better, players demand quality and respond in their favor.

CS2 acts as an benchmark for ownership to collectible games

# 05 Challenges, Evaluation & Insights

Core Challenges of Blockchain and Tokenization in Gaming Industry

## Technical & UX Friction

- **Scalability & Latency**  
Fast gameplay vs. Blockchain processing
- **Cost & Infrastructure**  
Transaction fees + network congestion + Less UX
- **UX Complexity**  
Wallet → Keys → Tokens/NFTs → Fees → Marketplace
- **Case Study: Ubisoft Quartz**
  - Crypto wallet required (third parties)
  - Key management
  - Registration delay

## Technical & UX Friction

- **New Security Attack Surfaces**
  - Wallets & private keys
  - Smart contracts & bridges
  - Validators & on-chain marketplaces
- **Traditional Gaming**  
Centralized servers & account security
- **Blockchain Gaming**  
Wallets + Smart Contracts + Bridges + Validators
- **Case Study: Ronin Exploit - 2022**  
≈ **\$625M stolen** (173,600 ETH + 25.5M USDC)

Blockchain does not simply improve gaming — it also introduces new layers of complexity, cost, and risk.



# Challenges, Evaluation & Insights

## Core Challenges of Blockchain and Tokenization in Gaming Industry

### 01 — NFT Ownership $\neq$ Asset Ownership

Owning the token does not mean owning the underlying game asset or IP

### 02 — Tokenization $\neq$ Interoperability

A transferable token does not mean it can be used across different games

### 03 — NFT Ownership $\neq$ Asset Ownership

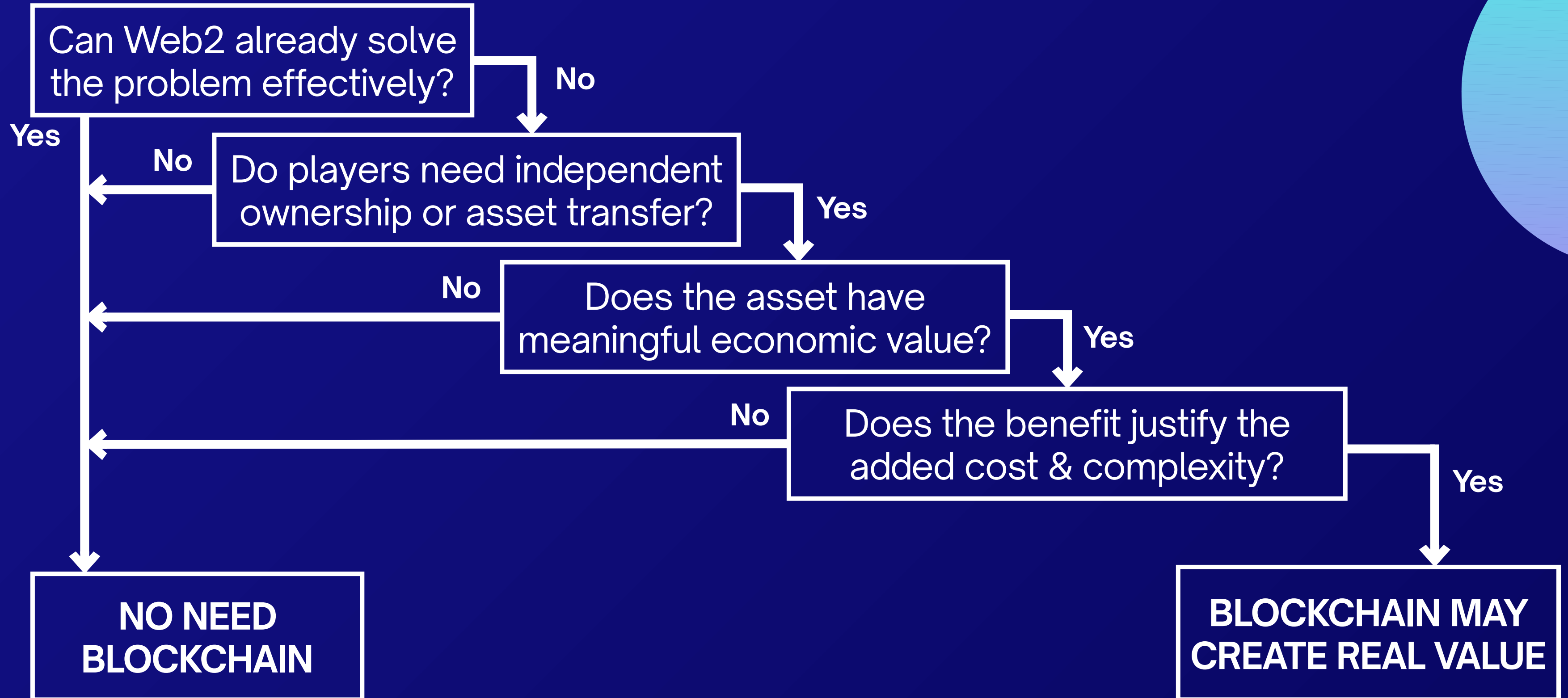
Blockchain shifts security and ownership responsibilities to wallets, keys, and smart contracts.

Blockchain can give players ownership of a token, but not necessarily the underlying asset, its utility, or cross-game usability.



# Challenges, Evaluation & Insights

Core Challenges of Blockchain and Tokenization in Gaming Industry



# Challenges, Evaluation & Insights

Core Challenges of Blockchain and Tokenization in Gaming Industry

**Gameplay First,  
Tokenization Second**

**Adopt Blockchain  
Selectively**

**Invisible Web3  
(UX first)**

**Design the Economy  
before the Token**

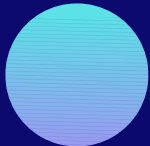
# Challenges, Evaluation & Insights

Core Challenges of Blockchain and Tokenization in Gaming Industry



**Gameplay First,  
Tokenization Second**

(1) Gameplay → (2) Player Demand  
→ (3) Economic Utility → (4) Token  
Necessity  
→ (5) Token Design



**Gameplay creates value**  
→ Fun & Engagement  
→ Competition/Creativity/Social Value  
**Tokenization supports it**  
→ Token = Supporting Utility  
→ Token ≠ Primary reason to play



**Design the Economy  
before the Token**



# Challenges, Evaluation & Insights

Core Challenges of Blockchain and Tokenization in Gaming Industry

**TOKENIZE WHEN →**

Rare • Valuable • Tradable • Player-driven

**DON'T TOKENIZE JUST BECAUSE →**

“Blockchain is available.”

**Adopt Blockchain  
Selectively**

**Invisible Web3  
(UX first)**

Blockchain should work behind the game  
— not get in the player's way.

# 06 Conclusion

“Does Blockchain actually fix Gaming?”

## Web2

### [Counter-Strike 2]

Valuable digital economies can arise from gameplay, status, and player demand — no blockchain required.

## Web3

### [Axie Infinity]

Token-driven incentives can create speculative dynamics; sustainability risks when monetary flows stop.

## What does

### blockchain adds?

Can provide verifiable ownership and control over tokenized assets — but ownership  $\neq$  playability or enduring demand.

**Keep the game fun first — tokens only where they solve a real problem.**

# Conclusion

The Future of Blockchain in Gaming

“The goal is not to make gaming  
more blockchain -  
but to make gaming better  
where blockchain adds value.”

# Thanks for Listening

Do You Have Any Questions?





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